[Re] Graph Edit Networks

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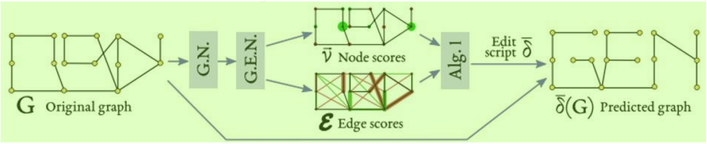
Reproduced: Paaßen et. al. (ICLR, 2021)

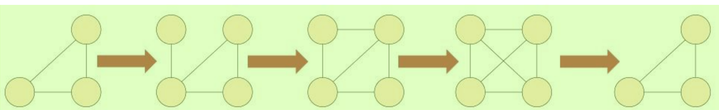
The reproduced paper [bpa21] proposes a novel output layer for graph neural networks (GNNs), the graph edit network called (GEN). This layer yields a sequence of graph edits: node insertions, node deletions, node replacements, edge insertions and edge deletions. These finite sequences of edits, also referred to as edit scripts are general enough to describe any graph-to-graph transformation and are not only very interpretable for humans, but also computationally efficient. These properties establish GENs as a useful tool for work in the domain of graph time series prediction under the Markovian assumption.



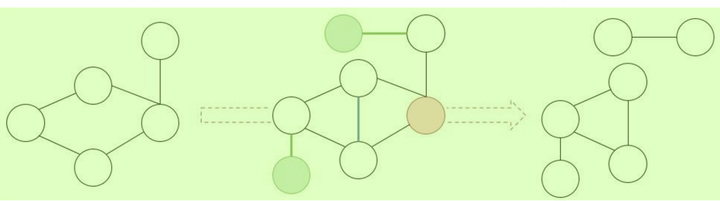
The authors empirically underpin their theoretical claims about GENs by showing that they perform well in a series of graph time-series prediction tests. They define several data generating processes (DGPs), from which the GEN attempts to learn the user-defined mapping functions. Our work reproduces these experiments and evaluates the suitability of the DGPs for the conclusions made in the original paper.

DGPs & Tests

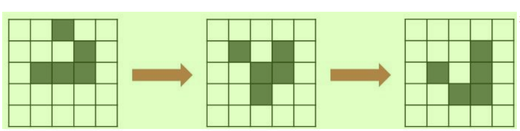
The paper evaluated several key claims on synthetic datasets, generated with data generating rules, interpreted in the following figures.



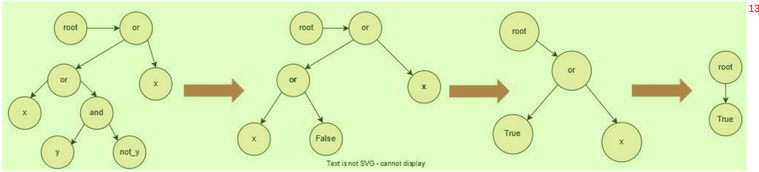
Graph cycles: A trivial set of user defined graph transformations without any logical rules.



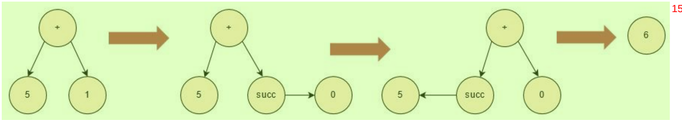
Degree rules: A set of rules for node & edge insertions. (nodes with degree ≥ 3 are deleted; noes with common neighbor are connected; where these do not apply, add a new neighboring node.)



Conway's Game of Life: A series based on node edits (editing Alive/Dead binary feature) on a lattice graph, adhering to Conway's Game of Life [Gar70].



Boolean Formulae: Random Boolean formulae, simplified with rules like $x \wedge \neg x \rightarrow \bot$, until tree cannot be simplified.

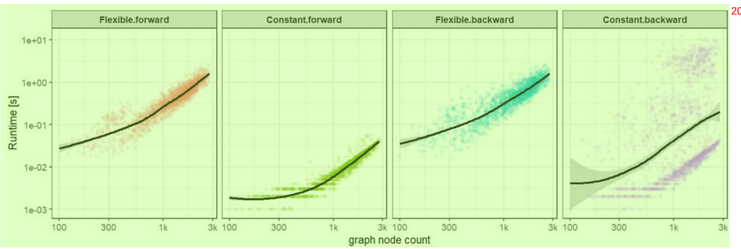


Peano Addition: Series of tree graphs, modelling addition problems via Peano's definition of addition ($m + succ(n) = succ(m) + n$ and $m + 0 = m$) using one-hot node features.

[Re] Original experimental claims

[bpa21] makes three experimental conclusions based on a series of experiments. ✓ symbols below denote a successful reproduction of results.

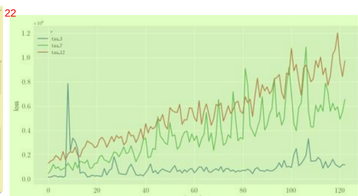
- ✓ GENs outperform baselines (Variational Graph Autoencoders [KW16] & Variational Graph Recurrent nets [HHN19]) on time-series prediction tasks, generated from Graph Cycles, Degree Rules & Game of Life DGPs.
- ✓ GENs achieve 100% accuracy on time-series prediction tasks on graphs, generated from Boolean Formulae & Peano Addition DGPs.
- ? The runtime of forward passes of a GEN scales sub-quadratically as the number of nodes in a graph increases. The runtime of training back-passes scales approximately linearly.



The runtime of graph scale dependence. Results for the backward passes are not consistent with the claim.

Pass direction	Edge filtering	Log-log linear slope fit
Forward	Flexible	1.38 ± 0.02
	Constant	1.31 ± 0.02
Backward	Flexible	1.30 ± 0.01
	Constant	1.69 ± 0.10

The runtime scaling evaluation was carried out on the ArXiv citation network [LKF05] with the graph variants generated by a different number of months considered.



While the focus here is speed, we have to note that the experiment in the original paper fails to optimize the criterion loss, thus casting doubt on the results.

Improving the experimental setup

The facts that graph-to-graph transitions are deterministic and that the space of unique graphs that may be generated by the DGPs is limited (as shown in the table below), arose some concern about the experimental setup: Does the solver generalize, or does it simply memorize the graph-to-graph transitions?

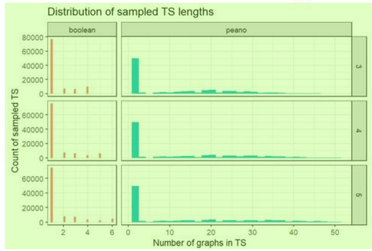
	Graph cycles	Degree rules	Game of life	Boolean formulae	Peano addition
# unique graphs	9	12346	2^{100}	10788	34353

We reproduced the claims (1) and (2) with the following adjustments:

- Increased cardinality of test time series (5 -> 100)
- Ensured no duplicates between train & test set
- Used established random graph generation methods (ie. Erdős–Rényi)

We noticed that most sampled graphs in the tree dynamic graph experiments were un-simplifyable, so constraints were relaxed here.

Results were generally worse, but still within a reasonable margin of error.



DGP	Prob. of sampling unique & simplifyable tree
Boolean formulae	0.13 ± 0.003
Peano addition	0.26 ± 0.002

Concluding reproduction notes

Our experimental results conclusively show that most of the claims in the original work hold. However, our work beyond the original paper emphasizes the need to pay attention to the experimental setup, especially when working with synthetically generated data, sampled from functions, rather than gathered from an external system.

References

[bpa21] Paaßen, Benjamin, et al. "Graph edit networks." In: International Conference on Learning Representations. 2020.

[Gar70] Martin Gardner. "The fantastic combinations of John Conway's new solitaire game of life." In: Sc. Am., 223:20–123, 1970

[KW16] Thomas N. Kipf and Max Welling. "Variational graph auto-encoders" In: Proceedings of the NIPS 2016 Workshop on Bayesian Deep Learning, 2016

[HHN19] Hajiramezanali, Ehsan, et. al. "Variational graph recurrent neural networks." In: Proceedings of the 32nd International Conference on Advances in Neural Information Processing Systems (NeurIPS 2019), pp. 10700–10710, 2019

[LKF05] Leskovec, Jure, Jon Kleinberg, and Christos Faloutsos. "Graphs over time: densification laws, shrinking diameters and possible explanations." In: Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining. 2005.